

UK HEAT SPLIT

Data sources and estimation methodology

Causeway Energies · causewaygt.github.io/uk-heatsplit · v4.0.0 · July 2026 · contact@causewaygt.com

Abstract

UK Heat Split is a public, automatically updated estimate of the weekly heating and cooling energy balance of Great Britain. Daily gas throughput to the low-pressure distribution network is decomposed into a temperature-sensitive space-heating component and a weather-independent baseline by segmented linear regression on population-weighted degree days, following the established gas-to-heat method of Watson et al. [1,2] and Sansom [3]. The trailing twelve-month space-heating estimate is calibrated against the Department for Energy Security and Net Zero (DESNZ) Energy Consumption in the UK (ECUK) end-use tables [4]. Non-gas fuels and space cooling, which lack daily operational feeds, are held at annual ECUK levels and modulated by degree days. Delivered service is derived from purchased energy using in-situ combustion efficiencies [5] and measured seasonal heat-pump performance factors [6,7]. Cooling demand in the unequipped building stock is estimated independently from overheating degree-hours above the CIBSE comfort threshold [8] and dwelling-stock statistics [9,10]. All emissions use DESNZ conversion factors [11]; all monetary values use the prevailing Ofgem price cap [12]. Every figure resting on author judgement is flagged; the complete source code and data are public. This document specifies each estimator, its inputs, its assumptions, and its uncertainty.

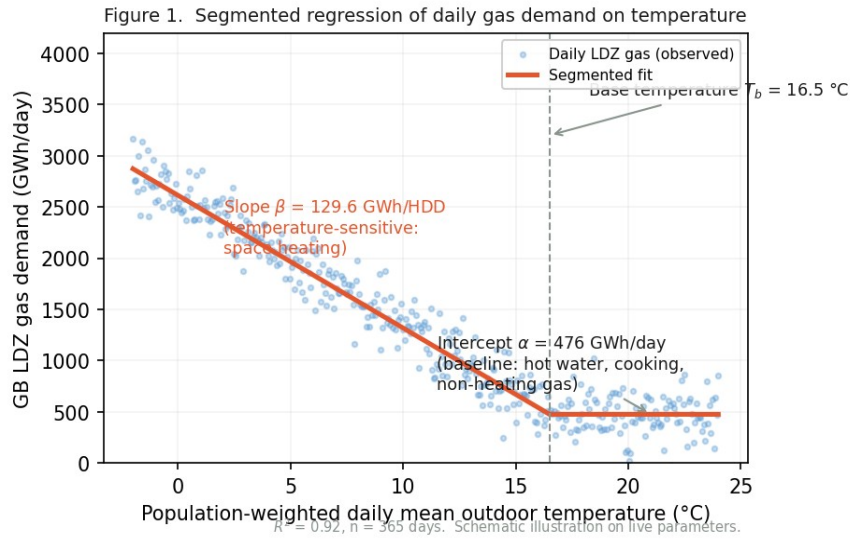
1. Conventions

1.1 Sourced quantities and author estimates

Two classes of number appear on the dashboard. Sourced quantities are traceable to a cited publication or a live public feed. Author estimates (marked † on the site) rest on Causeway Energies judgement where no authoritative figure exists; each is accompanied by a stated range and an invitation to challenge. The distinction is maintained throughout this document.

1.2 Purchased energy and delivered service

Two physical quantities are tracked and must not be conflated. **Purchased energy** is metered fuel or electricity at the point of supply. **Delivered service** is the useful heat or cooling supplied to the conditioned space after conversion. For a combustion appliance of efficiency $\eta < 1$, delivered service is less than purchased energy; for a heat pump of coefficient of performance $COP > 1$, delivered service exceeds purchased electricity because ambient or ground heat, never purchased, is harvested. The indigenous-supply and efficiency arguments on the site depend on this distinction.



2. Primary heating estimate: gas-to-heat regression

2.1 Data source

The primary feed is the National Gas Transmission operational-data REST API [13], from which the daily aggregate demand of the thirteen Local Distribution Zones (LDZ) is retrieved and summed. Publication identifiers are resolved from the API catalogue at runtime rather than hard-coded, so that a renamed publication produces a diagnostic rather than a silent error. The total National Transmission System (NTS) demand is retrieved separately for context. The LDZ aggregate is used in preference to the NTS total because the latter includes gas-fired power generation and directly connected industry, which are unrelated to building heat and would dominate the signal.

2.2 Quantifying the LDZ / NTS distinction

The distinction is material. In the current week the summed LDZ demand is of order 480 GWh/day at summer minimum, against an NTS total of order 1,000–1,400 GWh/day; the difference is principally power-station and industrial offtake carried on the transmission system, which does not track heating degree days. Using the NTS total would introduce a large, weather-insensitive term and depress the apparent temperature sensitivity of demand.

2.3 Temperature index

Outdoor temperature is represented by a population-weighted mean over twelve Great British urban centres (London, Birmingham, Manchester, Leeds, Glasgow, Newcastle, Bristol, Cardiff, Edinburgh, Southampton, Nottingham, Sheffield), with weights from Office for National Statistics mid-year population estimates. Hourly and daily 2 m air temperatures are drawn from the ERA5 reanalysis [14] via the Open-Meteo archive API [15]. Heating degree days at base temperature T_e are computed as $HDD = \max(0, T_e - \bar{T})$, with $\bar{T}$ the weighted daily mean; cooling degree days use $CDD = \max(0, \bar{T} - T_c)$. Bases are evaluated at 14.5, 15.5 and 16.5 °C for heating and 18 °C for cooling; the heating base is selected on goodness of fit.

2.4 Estimator

Daily LDZ demand G is modelled as a segmented (piecewise-linear) function of temperature, equivalent to an ordinary-least-squares regression on the degree-day transform:

$$G_i = \alpha + \beta \cdot HDD_i + \varepsilon_i$$

where α (GWh/day) is the weather-independent baseline attributed to water heating, cooking and other non-space-heating gas use; β (GWh per degree day) is the marginal space-heating response; and ε is a residual. The fitted space-heating component on day i is $\beta \cdot HDD_i$, and the baseline is α . This is the decomposition of Watson et al. [1,2], itself building on Sansom [3]; the

method is standard in the GB heat-demand literature. Current fitted parameters are $\alpha \approx 476$ GWh/day, $\beta \approx 130$ GWh/HDD, with base $T_e = 16.5$ °C over a 365-day window (Figure 1).

2.5 On the coefficient of determination

The regression returns $R^2 \approx 0.92$. The precise statement is that approximately 92% of the *variance* in daily LDZ demand across the window is accounted for by the degree-day term — not, strictly, 92% of the demand. The earlier informal gloss ("nine-tenths of the variation") is correct only in the variance sense and is stated as such here. Two caveats attach. First, because HDD and demand share a strong annual cycle, a high R^2 is expected and does not by itself validate the functional form. Second, R^2 is inflated by the seasonal range; the residual standard error (of order 180 GWh/day) is the more informative figure for daily uncertainty and is reported in the calibration diagnostics.

3. Calibration to national statistics

The regression yields a self-consistent daily signal but not an absolute national total. The trailing twelve-month integral of the fitted space-heating component is therefore reconciled against ECUK [4], which reports domestic and non-domestic energy consumption by end use and fuel, derived by DESNZ from the Digest of UK Energy Statistics (DUKES) national energy balance combined with the Cambridge Housing Model and the National Energy Efficiency Data-Framework. ECUK space-heating totals are themselves modelled estimates, weather-corrected to a long-run average using the same degree-day apparatus, not metered heat.

The calibration ratio $r = (\text{modelled 12-month space heat}) / (\text{ECUK anchor, GB-adjusted and weather-normalised})$ is monitored continuously. The publication gate is $|r - 1| \leq 0.10$. The current value is $r \approx 1.10$, i.e. the live model runs about 10% above the ECUK anchor — at the edge of tolerance, and reported openly on the site rather than silently rescaled. Candidate contributions to the residual include the treatment of non-domestic gas, the fixed base temperature, and weather-normalisation differences between the trailing window and the ECUK reference year.

Figure 2. Within-class centring (month $\times$ weekend) isolates the cooling anomaly

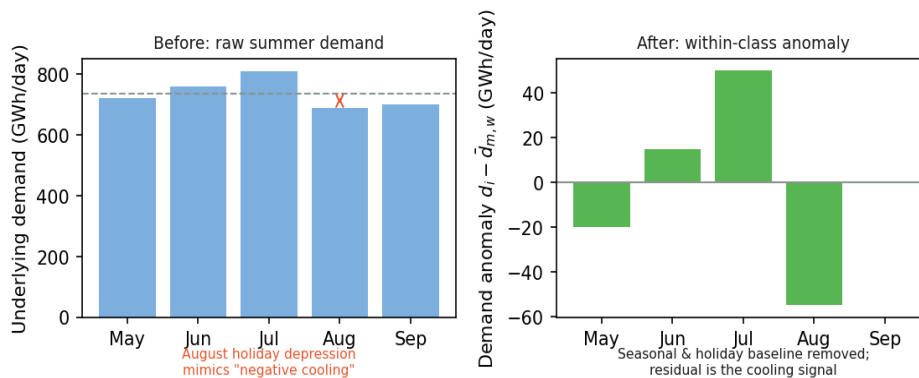


Figure 2. Within-class centring. Raw summer demand (left) is depressed in August by the holiday reduction in commercial and industrial activity, which a naive regression misreads as negative cooling. Subtracting the month $\times$ weekend-class mean (right) removes the seasonal and holiday baseline, leaving the cooling anomaly.

4. Cooling: the equipped fleet (observed)

4.1 Data source and reconstruction

Cooling delivered by installed plant is inferred from national electricity demand. Half-hourly demand is taken from the National Energy System Operator (NESO) historic demand data and rolling update files [16]. Metered demand understates true underlying demand by the quantity supplied behind the meter, principally embedded solar photovoltaics, which is largest precisely on the hot, bright days when cooling load peaks. Underlying demand is therefore reconstructed by adding NESO estimates of embedded solar and wind generation to metered demand before

analysis. Future-dated rows present in the update file, and days with other than 48 half-hourly values, are removed.

4.2 Within-class centring

Summer electricity demand carries a strong non-cooling structure: it falls during the August holiday period and differs between weekdays and weekends. Binning raw demand against CDD conflates this structure with the cooling response and can yield a spurious negative uplift. To isolate the cooling signal, each daily underlying-demand value d_i is expressed as an anomaly about the mean of its (calendar-month $\times$ weekend/weekday) class:

$$a_i = d_i - \bar{d}(m_i, w_i)$$

where $\bar{d}(m, w)$ is the mean underlying demand over all days sharing month m and weekend/weekday status w . This within-class centring (Figure 2) removes the seasonal and holiday baseline while preserving the temperature-driven excursion. The operation is a fixed-effects demeaning in the econometric sense; the term "demeaning" is avoided in public-facing copy in favour of "within-class centring" to prevent confusion with the ordinary meaning of the word.

4.3 Response curve and latent slope

The anomalies a_i are grouped into CDD bins; the mean anomaly per bin is the observed cooling response (Figure 4). A linear slope fitted to the low-CDD bins, where the fleet operates unconstrained, gives the latent response in GWh per CDD; extrapolating it defines the demand that the equipped fleet would meet in the absence of capacity or behavioural limits. Departure of the observed curve below this line at high CDD would indicate saturation. The current curve does not saturate — the highest bin shows the largest uplift — consistent with field evidence that the installed UK cooling stock is not capacity-constrained at present temperatures, and implying headroom rather than a ceiling.

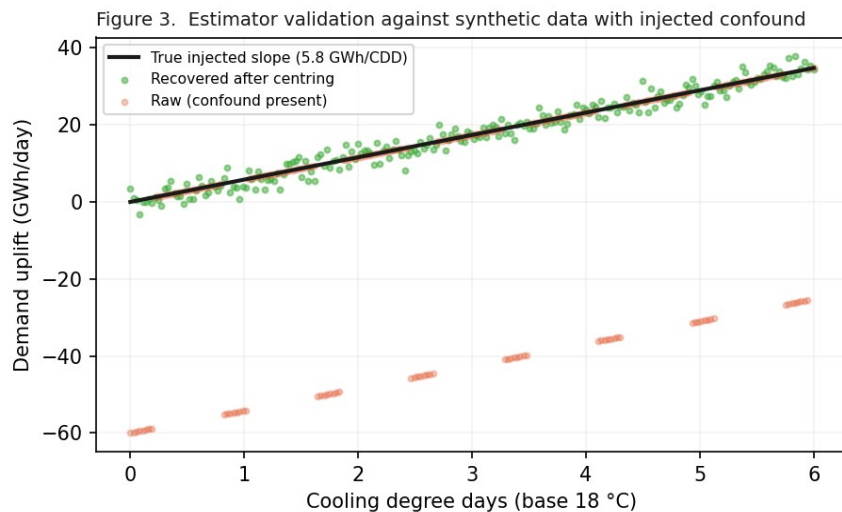


Figure 3. Estimator validation. A synthetic demand series was constructed with a known cooling slope and a holiday depression deliberately injected. The raw binning (orange) is biased downward by the confound; within-class centring recovers the true slope (green vs. black).

4.4 Validation

The centring estimator was validated before deployment on synthetic data in which a known cooling slope was combined with an injected holiday depression of comparable magnitude to that seen in real August demand (Figure 3). Naive binning of the synthetic series reproduced the spurious negative-uplift artefact; within-class centring recovered the injected slope to within sampling error. This confirms that the procedure removes the confound it is designed to remove without distorting the underlying signal.

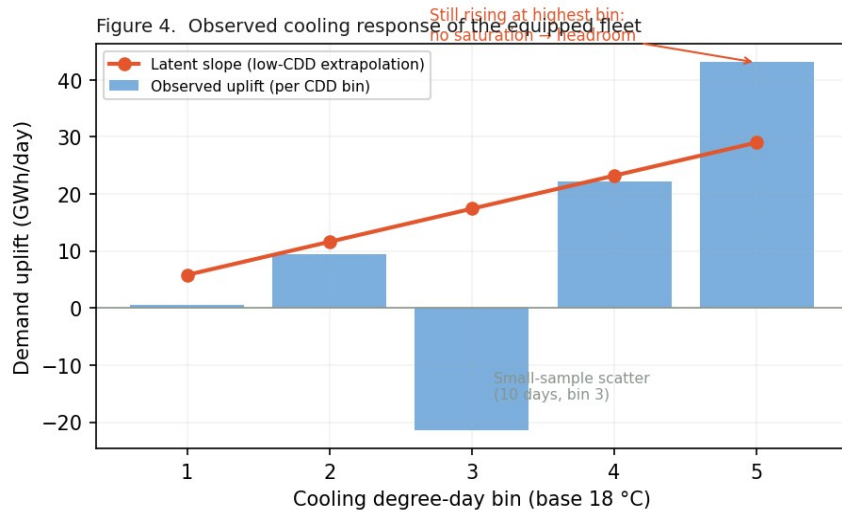


Figure 4. The live cooling response curve (this is the chart shown on the site). Bars are the mean demand uplift per CDD bin; the line is the latent low-CDD slope extrapolated. The negative third bin is small-sample scatter (ten days); the continued rise at the highest bin indicates no saturation.

4.5 Relationship to the billed cooling figure

The observed electricity response yields a weekly cooling electricity of order 150–230 GWh, whereas the ECUK-anchored figure used in the cost and carbon panels is of order 1,000–1,200 GWh. This order-of-magnitude discrepancy is stated on the site and left unreconciled pending a full cooling season. Both estimates may carry error. The observed figure captures only the marginal, weather-driven increment in national demand and will *understate* cooling to the extent that base-load cooling (data centres, process refrigeration, continuously conditioned commercial space) is weather-insensitive and absorbed into the class means. The ECUK-shaped figure, conversely, distributes an annual national cooling estimate across the year by CDD and may *overstate* a given summer week if the annual ownership or intensity assumptions are high, or if the CDD base temperature (18 °C) misrepresents the temperature at which conditioning actually begins. Reconciliation via a year-round demand regression with explicit heating, cooling, weekday and seasonal terms is the planned resolution.

5. Cooling: the comfort deficit (modelled)

The observed method sees only buildings that possess cooling. Since fewer than 5% of UK dwellings are air-conditioned [17], the electricity signal is structurally blind to the majority of the population experiencing summer overheating. This latent demand is therefore estimated independently, and explicitly as a stock-and-engineering calculation rather than an inference from the electricity data.

Overheating exposure is quantified as overheating degree-hours above the CIBSE operative-temperature threshold of 26 °C for occupied spaces [8], population-weighted, from the same hourly ERA5 series:

$$ODH_{26} = \sum_h \max(0, T_h - 26)$$

summed over hours h in the week. All hours are counted; an occupied-hours weighting (a candidate refinement, †) would reduce the total. The latent thermal demand is ODH_{26} scaled by the affected, unequipped dwelling stock and a per-dwelling thermal-response coefficient, plus an equivalent term for uncooled non-domestic comfort floor area derived from the Building Energy Efficiency Survey [18].

5.1 Uncertainty on the stock and response assumptions

The dominant uncertainties are the overheating-prevalence fraction and the thermal-response coefficients, all author estimates (†). Three prevalence scenarios are carried and never averaged: a low case of 11% of dwellings, from English Housing Survey self-reported overheating [9]; a high case of over one-half, from the Climate Change Committee assessment of dwellings at overheating risk [10]; and a central case of 25% (†). The per-dwelling response

(order 0.2 kWh per degree-hour †) and the non-domestic response (order 3 Wh m⁻² per degree-hour †, over ~190 Mm² †) are engineering judgements with a plausible factor-of-two spread each. These ranges bound the latent-demand estimate; they are independent of, and do not narrow, the observed-versus-ECUK discrepancy of Section 4.5, which concerns the equipped fleet rather than the deficit.

5.2 Delivery efficiency and the geocooling comparison

The central latent thermal load is expressed as the electricity required to meet it by two routes: passive or near-passive ground cooling at an effective COP of order 20 (†), and air-source vapour-compression at a seasonal EER of order 3.5. The resulting six-to-one ratio quantifies the electrical advantage of ground-coupled cooling. The seasonal-storage corollary — that heat rejected to an aquifer or borehole field in summer is recoverable in winter at a round-trip thermal efficiency of order 70% (literature range 50–80%) [19] — is shown in the dashboard's seasonal-mirror graphic with the loss represented explicitly.

6. Conversion, cost and carbon

6.1 In-situ efficiencies and performance factors

Combustion appliances are derated by in-situ, not nameplate, efficiencies: gas 0.835 and oil ~0.82 [5]. Heat pumps are credited with a seasonal performance factor of 3.24 for ground-source systems, from the Energy Systems Catapult in-situ monitoring of more than 1,700 installations [6]; the associated cold-weather stability of ground-source relative to air-source is documented in the Electrification of Heat demonstration [7]. These are the values underlying the delivered-service bars and the daily spark gap.

6.2 Cost

Retail costs use the prevailing Ofgem default-tariff cap unit rates [12], refreshed quarterly. The daily spark gap uses wholesale commodity prices only — National Gas System Average Price [13] and the Elexon market index [20] — divided by the respective efficiency or performance factor, and excludes network, policy and supply costs. Oil is priced from published heating-oil surveys (†).

6.3 Carbon

Combustion emissions use DESNZ (Defra) greenhouse-gas conversion factors [11], reported on a gross calorific value basis consistent with UK metered gas billing: natural gas 0.18296 kgCO₂e/kWh (2025), kerosene ~0.247, coal ~0.345. Electric routes use the live GB grid carbon intensity from the NESO Carbon Intensity API [21], as a seven-day mean. Bioenergy is counted at zero combustion emissions under the biogenic convention, excluding supply-chain emissions — a stated simplification. Because combustion factors are fixed by fuel chemistry while the electric factor tracks the decarbonising grid, the heat-pump rows fall over time and the combustion rows do not; the panel is designed to make this divergence legible.

7. Indigenous-supply share

The headline indigenous share is computed on a delivered-service basis: each unit of delivered heat or cooling inherits the UK-origin fraction of its energy input — gas ~38% (UK Continental Shelf share of supply, DUKES [22]), grid electricity ~75% (†, from generation-mix and interconnector data), harvested ambient and ground heat 100%. The last follows the Eurostat and DUKES convention of treating captured ambient heat as indigenous renewable supply [22,23]. Cooling's delivered multiple inherits the origin share of its input electricity; thermodynamic leverage is not treated as an energy origin. The purchased-energy-basis share is retained in the published data for continuity; it changes little under a geothermal shift because ground heat is never purchased, which is the substantive point the services basis is designed to expose.

8. Other panels

8.1 Non-gas fuels

Oil, bioenergy, solid fuel, heat networks and electric resistive heating lack daily operational feeds and are held at ECUK annual levels [4], modulated where appropriate by degree days. They move slowly and contribute little week-to-week variance.

8.2 Geothermal

The present geothermal contribution is anchored on the EGEN 2025 UK Country Update [24]: approximately 1.43 TWh/yr from an installed base of order 55,000 ground-source heat-pump units, plus of order 0.07 TWh/yr from deep, mine-water and open-loop district schemes (Gateshead, Eden, and others). The stated range is 1.4–2.0 TWh/yr (†); no national register or metering of ground-source output exists, and the Environment Agency’s operational-unit estimate is lower than the installed-unit figure, so this is the most weakly constrained quantity on the site. Forward bars for 2027 and 2031 are trend and scenario projections (†) informed by the EGEN Geothermal Market Report 2025 [25] (4,070 UK unit sales in 2025; four new large closed-loop systems; ~11 district systems in development) and the Climate Change Committee Seventh Carbon Budget pathway [26]; the 2050 bar is the Project InnerSpace/REA/ARUP ambition [27]. The European benchmark (2.55 million units delivering 88 TWh in 2025; Swedish sales of 26,785 units against UK sales of 4,070) is drawn directly from [25].

8.3 Northern Ireland

Northern Ireland operates separate gas and electricity systems and the all-island Single Electricity Market, and is not covered by the GB live feeds. Its heat is estimated annually from the Department for the Economy energy statistics and NISRA Continuous Household Survey [28], modulated by NI degree days. Oil heats just over 60% of NI dwellings and has no meter or daily feed; the resulting data gap is itself reported.

8.4 Whole-economy context (WHY HEAT?)

Four annual pies place heat beside transport and non-heat electricity across delivered services, spend, imported energy and emissions. Sourced anchors are ECUK 2025 final-energy and transport totals [4] and DESNZ provisional greenhouse-gas statistics by sector [29]; the service-level allocations (heat total ~630 TWh, spend, import and emission splits) are author derivations (†), deliberately rounded. Non-energy emissions are excluded; the view is of energy services, not a national inventory.

9. Automation, failure handling and reproducibility

A scheduled GitHub Actions workflow executes daily at 03:43 UTC: it retrieves each feed, fits the regressions, computes the balance, and commits a single JSON data file rendered by a static page. No authentication is required for any source; all are open data. The repository is public, so every computation behind every published number is inspectable. Each feed is wrapped so that a failure retains the last valid value and flags the affected panel; sources that publish on a lag are shown as provisional when their latest value is stale. Publication identifiers are resolved from source catalogues at runtime, and data units are inferred from magnitude, so that upstream renaming or unit changes surface as diagnostics rather than silent errors. Build failures raise an explicit notification, so that the absence of an alert is itself evidence that the pipeline is current.

10. Principal limitations

The following are the material limitations, stated plainly:

- All live figures are model estimates from temperature regression on operational data calibrated to annual statistics, not metered national heat.
- The gas-to-heat decomposition assumes a stable demand-temperature relation; it degrades in atypical weather and under price shocks, and cannot cleanly separate cooking and non-heating gas.

- The observed and ECUK-anchored cooling estimates differ by an order of magnitude and are not yet reconciled (Section 4.5).
- The comfort-deficit stock fractions and thermal-response coefficients carry a several-fold range (Section 5.1).
- Geothermal output is not nationally metered; the present-day bar is the most weakly constrained figure on the site (Section 8.2).
- Annual anchors (ECUK, DUKES) lag by roughly nine to fifteen months; the live layer is provisional until each annual release.

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*Every author estimate on the dashboard carries a dagger (†) and an invitation. Corrections and challenges to any figure, source or method in this document are welcomed at **contact@causewaygt.com**.*

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